Context Window Diseases and Cancer Risk: Neural Computation Cost as a Unifying Mechanism

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Abstract

We propose that psychiatric disorders classifiable as context window failures share molecular machinery with cancer. The direction of cancer risk depends on the failure mode. Mild context inefficiency, where the neural context-switching machinery operates at high cost per bit of uncertainty reduced, predicts elevated cancer risk. Severe context failure, where the machinery is disabled, predicts reduced cancer risk. This yields an inverted-U relationship between context dysfunction severity and cancer incidence.

The mechanism: neural computation requires topoisomerase-mediated DNA double-strand breaks for gene expression during learning and memory formation. Each context switch imposes a DNA damage cost independent of whether the switch produced useful knowledge. Disorders characterized by excessive context switching (ADHD, anxiety), context locking (PTSD, OCD), or context narrowing (depression) accumulate DNA damage at elevated rates relative to knowledge acquired. The chromatin remodeling genes that repair this damage, *ARID1A*, *KMT2C*, *SMARCA4*, *DNMT3A*, are the same genes whose somatic mutation drives cancer progression.

We formalize this using the bounded-context framework [1], deriving a mutation cost rate proportional to the inverse of effective channel capacity C_n. We classify psychiatric conditions by context window failure mode, map each to its known cancer risk profile, and show that the inverted-U pattern is consistent across sixteen conditions. We propose a three-tier clinical protocol: interrupt the thrash (somatic techniques), optimize the scheduler (cognitive therapy), tune the hardware (pharmacotherapy); with somatic mutation burden in chromatin remodeling genes from circulating free DNA as the molecular biomarker linking psychological intervention to cancer risk reduction. A meta-tier, molecular monitoring of chromatin remodeling gene mutations via liquid biopsy, operates on a different problem space: the escalation chain itself. Where the three treatment tiers optimize the context window, the meta-tier reads the state of the escalation hierarchy directly, detecting both accumulating damage and somatic modifications that have already altered the chain.

Twelve falsifiable predictions are stated.

1 Introduction

Learning has a physical cost. Neurons break their own DNA to learn.

Topoisomerase IIβ creates double-strand breaks in the promoter regions of immediate early genes during memory formation [5]. These breaks are necessary; without them, the chromatin cannot open fast enough for rapid transcriptional response. The cell repairs the breaks, but imperfectly. Each learning event leaves a residue of somatic mutation.

This is not damage from an external insult. It is the cost of computation. The cell breaks its DNA on purpose, reads the gene, repairs the break, and moves on. Over a lifetime, the repair errors accumulate.

The question is: does the *efficiency* of learning determine the rate of accumulation?

We argue that it does. Psychiatric disorders are classifiable by how they degrade the efficiency of the neural context window. The resulting excess DNA damage cost explains an otherwise puzzling pattern in the epidemiological literature: mild psychiatric disorders elevate cancer risk, while severe ones reduce it.

1.1 The cost per bit

In the bounded-context framework [1], an agent reduces its uncertainty about the world $H(W \mid K)$ by acquiring knowledge K through a channel with capacity C_n. The number of computational cycles required to reduce uncertainty by one bit is bounded below by $1/C_n$. Each cycle, in a biological neural network, involves gene expression changes that require DNA breaks.

The mutation cost per bit of uncertainty reduced is therefore:

$$\mu = \frac{\delta \cdot N_{\text{cycles}}}{\Delta H} \geq \frac{\delta}{C_n} \tag{1}$$

where δ is the per-cycle probability of a repair error (approximately 10^{-9} per base pair per repair event [6]), N_{cycles} is the number of computation cycles, and ΔH is the information gained.

An efficient learner has high C_n: each cycle reduces uncertainty productively. The DNA damage cost per bit is near the minimum δ/C_n.

A context window disorder reduces effective C_n without reducing N_{cycles}. The system computes at the same rate but learns less per cycle. The mutation cost per bit rises. Over decades, the difference compounds.

2 Context Window Failure Taxonomy

We classify psychiatric conditions by how they degrade the context window. The context window, in bounded-context theory [1], is the set of propositions simultaneously available for inference. It has a hard capacity bound C_n and supports four operations: load, hold, release, and switch.

Each operation can fail. The failure mode determines the computational waste pattern and, we argue, the cancer risk profile.

2.1 Premature eviction: context cannot hold

The window evicts propositions before they are consolidated. Each eviction triggers a full context reload, a new round of gene expression changes with associated DNA breaks—for a partial or zero information gain.

ADHD. The canonical premature eviction disorder. Attention shifts before consolidation completes. The system thrashes: high cycle count, low bits per cycle, elevated μ. Hyperfocus episodes represent the complementary failure; the window locks on a single context and refuses to release, which is costly only if the locked context is not productive.

A large Swedish registry study ($n \approx 4$ million) found ADHD associated with elevated overall cancer risk after adjustment for lifestyle confounders [7]. ADHD shares genetic variants with cell cycle

regulation and chromatin remodeling pathways, the same coupling channels identified in [3] as cancer-critical.

Traumatic brain injury. Physical damage to the context buffer. Elevated glioma risk is well-documented [8]. The mechanism in our framework: damaged tissue requires elevated repair activity, consuming DDR channel capacity that would otherwise catch somatic mutations.

2.2 Context lock: context cannot release

The window loads a context and cannot release it. New evidence cannot enter. The locked context replays repeatedly, each replay incurring the full DNA damage cost of gene expression, while the information gain is zero (the replay teaches nothing new).

PTSD. Traumatic context replays involuntarily. Each flashback activates the same neural circuits, the same gene expression programs, the same DNA breaks, in promoter regions already weakened by prior replay. Large Veterans Affairs studies show elevated cancer incidence in PTSD patients after adjustment for substance use and other behavioral confounders [9]. The standard explanation is cortisol-mediated immune suppression. Our framework adds a mechanism: the replay loop imposes a DNA damage cost with $\Delta H = 0$ per cycle.

OCD. Checking loops. The context window loads a threat assessment, cannot resolve it, clears, reloads, re-checks. Each cycle costs DNA breaks. No uncertainty is reduced because the check never terminates. Some epidemiological evidence supports mildly elevated cancer risk [10].

Addiction. The reward context overrides the scheduler. The window is permanently loaded with the reward-seeking program, suppressing context switches that would serve learning. Neural circuits associated with the addictive behavior are chronically active; associated DNA damage accumulates in those circuits specifically. Elevated cancer risk in addiction is well-documented, typically attributed to carcinogen exposure (smoking, alcohol). Our framework predicts elevated risk even for non-carcinogen addictions (gambling, behavioral addiction) through the context lock mechanism alone. Limited evidence is consistent with this [11].

2.3 Context narrowing: window contracts

The window's effective capacity shrinks. Fewer propositions are loaded per cycle. The system still computes, still breaks DNA, but each cycle processes a diminished context.

Depression. The context window collapses to a narrow negative prior. Rumination, the repetitive reprocessing of negative content, is computation with near-zero information gain. The system runs at full metabolic and DNA damage cost but acquires no new knowledge. Depression approaches a **near-zero-C_n state with nonzero compute cost**: as effective capacity $C_n^{\text{eff}} \to 0$, the mutation cost per bit $\mu \to \infty$. The system pays the full DNA damage cost of computation while acquiring almost nothing.

Meta-analyses consistently show elevated cancer risk in depression [12]. The standard explanation involves cortisol, inflammatory cytokines, and behavioral factors (reduced screening, poor diet). Our framework does not compete with these mechanisms. It adds a direct molecular channel: rumination-driven DNA damage in neurons and, via systemic stress signaling, in peripheral tissues.

Anxiety. The context window is too wide. Everything is loaded as a potential threat. The system evaluates an excessive number of propositions per cycle; most are irrelevant. The information gain per cycle is diluted by the noise. Mixed epidemiological evidence: some studies show mildly elevated cancer risk [13].

2.4 Context dissolution: boundaries fail

The window loses the ability to maintain coherent boundaries between contexts. Unrelated propositions bleed into each other. The system cannot maintain the separation between "self" and "other," between "current context" and "irrelevant context."

Schizophrenia. Context boundaries dissolve. Schizophrenia has *reduced* cancer risk in multiple large epidemiological studies [15, 16]. This is one of the most replicated and least explained findings in psychiatric epidemiology.

In our framework, the explanation is direct. Schizophrenia disables the context-switching machinery. Chromatin remodeling genes that are risk factors for schizophrenia overlap substantially with cancer-critical genes [17]. If these genes are already non-functional (from germline variants that cause schizophrenia), they cannot be *further* corrupted by somatic mutation to enable cancer. Moreover, cancer cells require functional chromatin remodeling to execute the gene expression reprogramming necessary for tumor progression, the switch from "normal cell context" to "invasive metastatic context." A system in which the context-switching machinery is broken cannot execute this transition.

This predicts that antipsychotic treatment, which partially restores context-switching function, should move cancer risk back toward baseline. Evidence supports this: antipsychotic use in schizophrenia is associated with increased cancer risk relative to unmedicated patients [18]. The standard explanation is metabolic side effects. Our framework offers a mechanistic alternative: restoring the chromatin remodeling machinery re-enables the context switches that cancer requires.

2.5 Context index failure: retrieval breaks

The window can still hold propositions, but the retrieval system that selects which propositions to load is corrupted. The wrong context is loaded, or no context can be loaded at all.

Alzheimer's disease. The context index is destroyed. Like schizophrenia, Alzheimer's has *reduced* cancer risk. The inverse comorbidity between Alzheimer's and cancer is among the most robust findings in epidemiology [19, 20]. The shared molecular basis centers on TP53: activation of p53 kills cells (prevents cancer, promotes neurodegeneration); loss of p53 lets cells survive (enables cancer, may protect neurons) [21]. In our framework: the same checkpoint that decides "this cell's context is corrupted, kill it" is the gate between the neurodegeneration attractor and the cancer attractor. They are opposite failure modes of the same context-integrity check.

2.6 Context oscillation

The window oscillates between incompatible contexts, spending resources on the transition without productive use of either.

Bipolar disorder. Context frames oscillate between manic (window too wide, loading everything, acting on incomplete evaluation) and depressive (window collapsed, loading nothing new). Both phases are computationally expensive. Mixed evidence: some studies show mildly elevated cancer risk [14].

2.7 Context rigidity

The window loads a context and holds it with excessive tenacity. Incoming evidence that does not fit the current frame is rejected rather than integrated. The system functions but at reduced

effective capacity: fewer propositions are considered per cycle because the acceptance filter is too narrow.

Autism spectrum. The context window is rigid: strong priors resist updating, sensory contexts that neurotypical systems filter are loaded at full resolution, and transitions between contexts are costly. The cancer risk profile is mixed [22], consistent with a heterogeneous severity spectrum—mild rigidity elevates cost per bit (the system over-processes), while severe rigidity reduces context-switching frequency (fewer cycles, less DNA damage).

2.8 Consolidation failure

The window operates during waking hours but the offline consolidation process, sleep, fails to compress episodic context into durable semantic structure. The proposition count n drifts toward the maintenance crossover n^* because compression cannot keep pace with acquisition.

Insomnia. Sleep consolidation is the brain's primary mechanism for reducing maintenance load (Section 8). Chronic insomnia degrades this process. The context window accumulates uncompressed propositions, each requiring active maintenance. Elevated cancer risk in chronic insomnia is consistent with the framework: the maintenance cost of unconsolidated knowledge drives excess chromatin remodeling demand [23].

2.9 Context initiation failure

The window cannot initiate a new context. The motor and dopaminergic circuits that translate a loaded context into action degrade, so the system loads contexts but cannot act on them. The stalled initiation loop consumes resources without producing behavioral output.

Parkinson's disease. The dopaminergic system that gates context-to-action translation degrades. Unlike schizophrenia, where the context-switching machinery itself is broken, Parkinson's preserves the ability to load contexts but impairs the ability to execute on them. Cancer risk is reduced for most solid tumors [24]. The framework explains this through a shared mechanism: the same dopaminergic circuits that fail in Parkinson's are required for the rapid gene-expression reprogramming that tumor cells use to switch context during invasion and metastasis.

2.10 Reduced baseline capacity

The context window has a constitutively smaller capacity from development. Fewer propositions can be loaded and maintained. The system operates within its reduced bounds—it does not thrash because it never attempts loads beyond its capacity.

Down syndrome. Trisomy 21 reduces effective context capacity constitutively. Solid tumor risk is markedly reduced [25]; the extra copy of chromosome 21 encodes several tumor suppressors, but in our framework the deeper mechanism is that the reduced context-switching demand means less chromatin remodeling wear. (Leukemia risk is elevated, consistent with the immune channel operating under different constraints.)

Intellectual disability (other causes). Germline variants that reduce cognitive capacity also reduce the demand on chromatin remodeling machinery. Some evidence supports reduced cancer risk in this population [26], consistent with the framework: lower context-switching load means fewer activity-dependent DNA breaks in the genes that cancer requires.

3 The Inverted-U Prediction

The taxonomy yields a structural prediction. Cancer risk is not monotonically related to context dysfunction severity. It follows an inverted U.

Definition 1 (Context dysfunction severity)**.** For an agent with baseline channel capacity C_n^*, context dysfunction severity s is defined as:

$$s = 1 - \frac{C_n^{\text{eff}}}{C_n^*} \tag{2}$$

where C_n^{eff} is the agent's effective channel capacity under the disorder. $s = 0$ for healthy function; $s = 1$ for complete context machinery failure.

Proposition 1 (Inverted-U cancer risk)**.** *Let $R(s)$ be cancer risk as a function of context dysfunction severity. Then $R(s)$ has an interior maximum: there exists $s^* \in (0, 1)$ such that $R(s^*) > R(0)$ and $R(s^*) > R(1)$.*

Remark (Argument)**.** The mutation accumulation rate from neural computation is:

$$\dot{M} = \delta \cdot f(s) \cdot g(s) \tag{3}$$

where $f(s)$ is the computation rate (cycles per unit time) and $g(s)$ is the fraction of cycles that produce non-productive DNA breaks (breaks not offset by adaptive benefit).

At $s = 0$ (healthy): f is moderate, g is low (most computation is productive). $\dot{M}$ is at baseline.

At intermediate s (ADHD, depression, PTSD): f remains high (the system still computes actively, even more so in anxiety and rumination), but g is high (most cycles are wasted). $\dot{M}$ is elevated.

At $s = 1$ (schizophrenia, severe Alzheimer's): f drops because the computation machinery is disabled. g is undefined because the machinery does not produce coherent cycles. $\dot{M}$ drops below baseline. Moreover, cancer cells themselves require functional context-switching machinery to progress; disabling it at the germline level is tumor-suppressive.

Therefore $\dot{M}$ (and consequently R) rises from $s = 0$ to intermediate s, then falls to $s = 1$, producing an interior maximum.

The available epidemiological evidence is consistent with this prediction. Table 1 maps sixteen conditions to their context failure mode, dysfunction severity, and observed cancer risk.

The table divides cleanly. Every condition with moderate dysfunction severity and intact computation machinery shows elevated cancer risk. Every condition with severe dysfunction or disabled computation machinery shows reduced cancer risk. The boundary falls where the machinery transitions from "working hard but inefficiently" to "broken."

4 The Shared Molecular Machinery

The context window framework predicts that psychiatric context disorders and cancer should share molecular machinery. They do.

Table 1: Context window disorders and cancer risk. Severity is qualitative (L = low, M = moderate, H = high). Cancer risk direction from meta-analyses and large registry studies.

Condition	**Failure mode**	**Severity**	**Cancer risk**	**Key reference**
ADHD	Premature eviction	L–M	↑	[7]
TBI	Buffer damage	L–M	↑	[8]
PTSD	Context lock	M	↑	[9]
OCD	Check loop	L–M	↑ (mild)	[10]
Addiction	Reward lock	M	↑	multiple
Depression	Context narrowing	M	↑	[12]
Anxiety (GAD)	Window too wide	L–M	↑ (mild)	[13]
Bipolar	Context oscillation	M	↑ (mild)	[14]
Autism	Rigid context	L–H	mixed	[22]
Insomnia	Consolidation failure	L–M	↑	[23]
Autoimmune	Context mis-indexing	M	↑	[43]
Schizophrenia	Context dissolution	H	↓	[15, 16]
Alzheimer's	Index destruction	H	↓	[19, 20]
Parkinson's	Context initiation	H	↓ (most)	[24]
Down syndrome	Reduced capacity	H	↓ (solid)	[25]
Intellectual disability	Reduced capacity	H	↓ (some)	[26]

4.1 Chromatin remodeling: the read/write heads

The genes that control chromatin accessibility, determining which genes are readable in which cellular context, are mutated in both psychiatric disorders and cancer:

- *ARID1A*: mutated in 6–8% of all cancers; germline variants associated with intellectual disability [27].
- *KMT2C*/*KMT2D*: among the most frequently mutated genes in cancer; Kabuki syndrome (germline *KMT2D* loss) involves intellectual disability [28].
- *SMARCA4*: mutated in lung, ovarian, and other cancers; Coffin-Siris syndrome (germline loss) involves intellectual disability [29].
- *DNMT3A*: mutated in AML; germline variants associated with Tatton-Brown-Rahman syndrome (overgrowth and intellectual disability) [30].

These are not coincidences. In the bounded-context framework, chromatin remodeling genes are the molecular implementation of context switching. They decide which genes are accessible (loaded into the context window) in a given cell state. When they fail by somatic mutation, the cell loses context control: it expresses genes that do not belong in its current state. That is cancer. When they fail by germline mutation, the organism loses context control at the neural level: it cannot efficiently switch cognitive contexts. That is intellectual disability.

The shared machinery explains both the elevated cancer risk in moderate context disorders (chromatin remodeling genes are overworked and accumulate somatic damage) and the reduced cancer risk in severe ones (germline loss of these genes prevents the context switches cancer requires).

4.2 TP53: the context integrity checkpoint

TP53 is the most frequently mutated gene in cancer [31] and a central node in the coupling-channel framework [3]. It functions as a context integrity check: if a cell's DNA is too damaged to maintain faithful gene expression context, p53 activates apoptosis.

In our framework, p53 is the *kill switch* for context window corruption. Its dual role in cancer (loss enables tumor growth) and neurodegeneration (activation kills neurons) is not a paradox. It is the same function, context integrity enforcement, applied in two domains where the optimal threshold differs.

Cancer patients need p53 active (to kill context-corrupted cells). Alzheimer's patients need p53 less active (to preserve neurons whose context is degrading with age). The inverse comorbidity between cancer and Alzheimer's [19] reflects the organism's position on a single axis: how aggressively it enforces context integrity.

4.3 The immune system as somatic hypermutation

The immune system stores knowledge in DNA through somatic hypermutation in B cells. This is not a metaphor for the context window framework. It is a literal implementation. The immune system:

1. Encounters an antigen (new evidence).
2. Mutates its own DNA to generate antibody variants (context switching at the molecular level).
3. Selects the variant that best matches (uncertainty reduction).
4. Stores the result as memory B cells (knowledge encoded in DNA).

Autoimmune diseases are context indexing failures: the retrieval system misidentifies self as threat, loading the wrong context. Autoimmune diseases are associated with elevated cancer risk [43], consistent with our framework: a context retrieval system that loads the wrong targets damages tissue through chronic inflammation and imposes DNA damage costs on cells that are not the intended target.

5 Clinical Protocol

If context-window efficiency is mechanistically upstream of somatic mutation accumulation, then treating context disorders is cancer prevention. The standard oncology framing treats psychiatric comorbidities as quality-of-life concerns. Our framework reframes them as disease-modifying targets.

We propose a three-tier treatment protocol ordered by intervention speed, cost, and side-effect profile. A meta-tier, molecular monitoring, operates on a different problem space entirely: not the context window but the escalation chain itself, detecting both accumulating repair burden and somatic modifications that have already altered the hierarchy.

5.1 Tier 1: Interrupt the thrash

Objective: Stop the runaway context loop immediately. Reduce N_{cycles} per unit time by providing a manual interrupt.

Interventions:

- **Somatic techniques** (EFT/tapping, grounding exercises, box breathing). These provide a hardware interrupt: when the cognitive context window is thrashing, a somatic stimulus forces attention to the body, evicting the runaway cognitive loop. EFT produces a 24% cortisol reduction in a single session [36]. Cortisol is a downstream marker; the upstream event is the interrupted context loop.
- **Bilateral stimulation** (EMDR eye movements, bilateral tapping). Forces alternating hemispheric activation, which disrupts the locked replay of a single context (PTSD mechanism).
- **Vagal tone exercises** (cold exposure, slow exhalation). Activates the parasympathetic branch, shifting the system from threat-context lock to rest-and-digest context.

Cost: Zero. **Latency:** Seconds to minutes. **Side effects:** None.

The rationale for Tier 1 as first-line: a thrashing system cannot be optimized while it is thrashing. The first intervention must halt the runaway process. Attempting cognitive restructuring (Tier 2) or pharmacotherapy (Tier 3) on a system in active thrash is analogous to optimizing a scheduler on a CPU that is currently deadlocked. Kill the runaway process first.

5.2 Tier 2: Optimize the scheduler

Objective: Retrain the context-switching rules so that future computation is more efficient. Increase C_n^{eff} without pharmacological intervention.

Interventions:

- **Cognitive behavioral therapy (CBT).** Explicitly rewrites the rules governing context selection. "When I notice thought pattern X, I switch to evaluation Y instead of loop Z." This is literally scheduler optimization.
- **Mindfulness-based stress reduction (MBSR).** Trains voluntary control over context loading and release. The practice of "noticing without engaging" is training the ability to observe a context candidate without loading it into the active window.
- **Acceptance and commitment therapy (ACT).** Separates context content from context switching. The content of an anxious thought matters less than whether the system loads, processes, and releases it efficiently.

Cost: Time (weeks to months). **Latency:** Weeks. **Side effects:** Minimal.

Mindfulness practice has documented effects on telomere maintenance [37] and inflammatory markers [38]. In our framework, these are downstream consequences of reduced context-switching waste: fewer unnecessary DNA breaks, less repair demand, less telomere erosion from replicative stress in repair-active tissues.

5.3 Tier 3: Tune the hardware

Objective: Pharmacologically modify the system parameters when Tiers 1 and 2 are insufficient. Modulate neurotransmitter systems to reduce loop gain (SSRIs), improve context efficiency (stimulants), or release context locks (beta-blockers).

Interventions and predicted cancer impact:

Table 2: Pharmacotherapy and context-window interventions with predicted cancer impact. Protective interventions improve context efficiency (bits per cycle); non-protective interventions reduce distress without improving efficiency or restore disabled machinery.

Drug class	Context effect	Predicted impact	Evidence
SSRIs	Reduce rumination loop gain	Protective	Reduced CRC risk [39]
Stimulants (ADHD)	Improve context efficiency	Protective	No excess risk [40]
Beta-blockers	Release threat context lock	Protective	Reduced metastasis [41]
Mindfulness (drug-free)	Reduce unnecessary switches	Protective	Telomere preservation [37]
Benzodiazepines	Reduce distress but impair formation	Not protective	Mixed [42]
Antipsychotics	Restore context machinery	Risk-neutral or harmful	Increased risk vs. unmedicated

The critical distinction: interventions that improve context efficiency (more bits per cycle) are predicted to be protective. Interventions that reduce distress without improving efficiency (benzodiazepines) are not. Interventions that restore disabled machinery (antipsychotics) may increase risk by re-enabling the context switches cancer requires.

5.4 Meta-tier: Monitor and repair the escalation chain

The three treatment tiers operate on the context window, reducing waste, improving efficiency, tuning parameters. This tier operates on a different problem space: the escalation chain itself. It reads the state of the molecular hierarchy directly, detecting both accumulating repair burden and somatic modifications that have already altered the chain. It is simultaneously diagnostic (has the escalation chain been damaged?) and therapeutic (can we intervene before the damage cascades?).

Objective: Verify that psychological intervention is reducing somatic mutation accumulation at the molecular level.

Circulating free DNA (cfDNA) from liquid biopsy can detect somatic mutations in chromatin remodeling genes (*ARID1A*, *KMT2C*, *SMARCA4*, *DNMT3A*) without tissue biopsy [32]. If the framework is correct, effective treatment of a context window disorder should reduce the accumulation rate of somatic mutations in these specific genes.

The proposed biomarker panel: somatic variant allele frequency in *ARID1A*, *KMT2C*, *KMT2D*, *SMARCA4*, *DNMT3A*, *TET2*, *SETD2*, and *CREBBP* from serial cfDNA draws at 6-month intervals.

A positive result, declining chromatin mutation VAF in patients whose context disorder is effectively treated, would constitute direct evidence that psychological intervention modifies cancer risk through a molecular mechanism, not merely through behavioral or hormonal intermediaries.

6 Falsifiable Predictions

1. **Inverted-U curve.** Across a severity spectrum of context window disorders, cancer risk peaks at moderate severity and decreases at both extremes (healthy and severely impaired).

2. **Chromatin specificity.** The elevated somatic mutation burden in moderate context disorders is concentrated in chromatin remodeling genes, not uniformly distributed across the genome.

3. **ADHD treatment is cancer prevention.** Stimulant medication for ADHD, by improving context efficiency, reduces somatic mutation accumulation rate in chromatin remodeling genes. Testable via cfDNA in treated vs. untreated ADHD cohorts.

4. **Depression treatment slows chromatin mutation.** Effective antidepressant treatment (defined as remission of rumination) reduces the rate of somatic mutation in chromatin remodeling genes. Ineffective treatment (symptom persistence) does not.

5. **Antipsychotics restore cancer risk.** Antipsychotic treatment in schizophrenia increases cancer risk relative to unmedicated schizophrenia, specifically through restoration of chromatin remodeling function. The effect should correlate with degree of cognitive improvement.

6. **Meditation is measurably protective.** Long-term meditation practitioners show lower somatic mutation burden in chromatin remodeling genes than age-matched controls, after controlling for lifestyle factors.

7. **Benzodiazepines are not protective.** Despite reducing anxiety symptoms, benzodiazepines do not reduce somatic mutation accumulation because they impair context formation without improving context efficiency.

8. **Non-carcinogen addictions carry cancer risk.** Behavioral addictions (gambling, internet, gaming) without chemical carcinogen exposure show elevated somatic mutation burden through the context lock mechanism alone.

9. **Chromatin mutations are increasing disproportionately.** The fraction of tumors with chromatin remodeling gene mutations is increasing faster than mutations in other gene classes over the last two decades, independent of improved detection.

10. **CHIP onset is shifting younger.** Clonal hematopoiesis driven by *DNMT3A*, *TET2*, and *ASXL1* is detectable at younger ages in post-smartphone cohorts than in age-matched historical controls.

11. **Early-onset cancers are enriched for chromatin mutations.** The cancers driving the early-onset trend (colorectal, breast, pancreatic in adults under 50) show higher rates of chromatin remodeling mutations than late-onset counterparts, after controlling for stage and subtype.

12. **Information exposure predicts chromatin VAF.** Context-switching load (measurable via smartphone usage data, notification frequency, or occupational classification) correlates with somatic mutation burden in chromatin remodeling genes from cfDNA, after controlling for age and carcinogen exposure.

7 Relationship to Prior Work

Psychoneuroimmunology. The field of psychoneuroimmunology has long studied how psychological states affect immune function and cancer outcomes [33]. Our contribution is mechanistic specificity: not "stress causes cancer through immune suppression" but "context-switching waste causes somatic mutation in the specific genes that control chromatin accessibility, through the specific mechanism of topoisomerase-mediated DNA breaks."

Activity-dependent DNA damage. Madabhushi et al. [5] demonstrated that neuronal activity causes targeted DNA double-strand breaks. Subsequent work showed that these breaks accumulate with age and correlate with neurodegenerative disease [34]. We extend this to cancer risk: if the break rate depends on computational efficiency, then disorders that reduce efficiency increase the break-to-knowledge ratio.

The Waddington connection. Waddington's genetic assimilation [35] showed that environmentally induced phenotypes can become genetically encoded. Our framework provides a mechanism: the epigenetic changes induced by experience (through chromatin remodeling during learning) are not merely reversible marks. They are accompanied by somatic mutations from imperfect repair of activity-dependent breaks. The mutations are permanent. Culture does not merely select on existing genetic variation. It generates new somatic variation through the physical cost of knowledge acquisition.

Coupling channels and cancer. In the companion paper [3], we showed that the number of coupling channels disrupted predicts cancer survival across 73,593 patients. The six coupling channels include chromatin remodeling (ChromatinRemodel) and DNA methylation (DNAMethylation) as the two channels added in the expanded eight-channel system. These are precisely the channels that the present paper identifies as the shared molecular substrate between context disorders and cancer.

Bounded context and intelligence. In [2], we derived the structural invariants of intelligence from bounded context, showing that evaluation-driven descent and encoding permanence are necessary properties of any system that reduces uncertainty under bounded resources. The present paper adds a biological cost to that descent: each step of evaluation-driven descent, each cycle of the intelligence loop, has a DNA damage cost. Intelligence is not free. Its cost is somatic mutation.

8 The Crossover: When Maintenance Exceeds Acquisition

Knowledge acquisition has diminishing returns. The marginal bit costs more as the knowledge base grows; the effective cost of the n-th proposition scales as $O(\log n)$ due to interference, retrieval complexity, and the need to integrate with existing knowledge.

Maintenance cost grows faster. Keeping n propositions coherent requires that each proposition remain accessible for potential interaction with others. The DNA damage cost of context maintenance, the ongoing chromatin remodeling required to keep gene expression programs responsive, scales with the number of context switches the system must service. In the worst case this is $O(n^2)$: any proposition might need to interact with any other.

These curves cross at some critical knowledge load n^*.

Definition 2 (Context maintenance crossover)**.** The crossover point n^* is the knowledge load at which the DNA damage cost of maintaining coherent access to n propositions equals the DNA

damage cost of acquiring the next proposition:

$$\delta \cdot c_{\text{maintain}}(n^*) = \delta \cdot c_{\text{acquire}}(n^*) \tag{4}$$

where $c_{\text{maintain}}(n) \sim n^2$ and $c_{\text{acquire}}(n) \sim \log n$.

Before n^*, learning is cheap relative to maintenance. After n^*, every new bit costs more in maintenance-driven DNA damage than it contributes in uncertainty reduction. The system must begin discarding propositions to stay below n^*, or the repair machinery will be overwhelmed.

8.1 The brain as first-line change management

The brain's primary strategy for staying below n^* is aggressive compression:

1. **Sleep consolidation.** Episodic memories (high n, high maintenance cost) are compressed into semantic memories (lower n, shared structure, amortized maintenance). This is the graph factoring operation from [1]: extracting shared structure to reduce the proposition count without losing information.

2. **Strategic forgetting.** Propositions that have not been accessed recently are evicted. This is the Belady replacement policy operating on the biological context window: evict the proposition whose next use is farthest in the future.

3. **Chunking.** Multiple propositions are compressed into a single proposition with internal structure (a "chunk"). This reduces n while preserving retrievability. Expert performance is characterized by larger chunks: a chess master stores a board position as one proposition where a novice stores 32.

4. **Habit formation.** Frequently accessed proposition chains are compiled into automatic routines that no longer require context-window loading. This is the substrate transition from [4]: when a computation becomes deterministic, it is removed from the uncertainty-reducing system and encoded as fixed procedure, reducing maintenance load on the active context.

These mechanisms keep n below n^* for decades. But they can be overwhelmed:

- **Chronic stress** loads threat propositions that cannot be compressed (because they are unpredictable) or evicted (because they are flagged as survival-relevant). They persist in active context, raising n toward n^*.

- **Trauma** creates propositions that cannot be integrated (they contradict the existing knowledge structure) or forgotten (they are tagged as maximally important). They occupy permanent slots in the context window.

- **Information overload** (a modern phenomenon) pushes the acquisition rate above the consolidation rate. Propositions accumulate in working memory faster than sleep can compress them. The maintenance cost rises while consolidation lags.

When the brain cannot hold the line, the biological context space absorbs the overflow. Chromatin marks accumulate beyond what the maintenance machinery can service. DNA repair backlogs grow. The read/write heads, the chromatin remodeling genes, wear out.

8.2 Aging as maintenance overload

The crossover point n^* is not fixed. It depends on the efficiency of the repair machinery, which degrades with age. As repair fidelity declines, n^* decreases: the system can maintain coherent access to fewer propositions before the maintenance cost overwhelms repair capacity.

This reframes biological aging as context maintenance overload. The organism does not degrade because time passes. It degrades because the accumulated knowledge load eventually exceeds the capacity of the molecular maintenance machinery, and the resulting DNA damage in chromatin remodeling genes progressively corrupts the context-switching apparatus.

The observation that cancer incidence rises sharply with age is consistent: once $n > n^*$ is persistent, the chromatin remodeling machinery is chronically overloaded, somatic mutations in context-switching genes accumulate, and the probability that a cell loses context control (becomes cancerous) rises.

The observation that cognitive decline precedes many cancers is also consistent: the brain crosses n^* first (because it has the highest context-switching demand), begins losing propositions, and the systemic maintenance overload follows.

9 The Information Age Prediction

If the mechanism proposed here is correct, then a society-level increase in context-switching load should produce a society-level increase in somatic mutation burden, concentrated in chromatin remodeling genes specifically.

The last twenty years have seen an unprecedented increase in information exposure. Smartphones, social media, and continuous connectivity have dramatically increased the rate of context switching in daily life. A pre-smartphone adult might execute dozens of major context switches per day. A smartphone user executes hundreds: checking notifications, switching between apps, processing social media feeds, responding to messages. Each switch loads a new cognitive context, triggers gene expression changes in active neural circuits, and imposes the associated DNA damage cost.

9.1 CHIP as the canary

Clonal hematopoiesis of indeterminate potential (CHIP) is the age-related accumulation of somatic mutations in blood stem cells. CHIP is detected in approximately 10% of people over 65 and increases with age [44]. It predicts elevated risk of both hematologic malignancy and cardiovascular disease.

The three most common CHIP driver mutations are:

1. *DNMT3A*: a DNA methyltransferase. Chromatin/epigenetic machinery.
2. *TET2*: a methylcytosine dioxygenase. Chromatin/epigenetic machinery.
3. *ASXL1*: a chromatin remodeling factor. Polycomb group regulation.

All three are chromatin remodeling genes. Together they account for approximately 80% of CHIP mutations [45]. This is not what a uniform mutagenesis model predicts. If CHIP were driven by random replication errors, the mutations would be distributed across the genome in proportion to

gene size and replication timing. They are not. They are concentrated in the genes that control chromatin accessibility.

Our framework explains this concentration. Blood stem cells reside in the bone marrow, are innervated by sympathetic fibers, and respond to systemic stress signaling (cortisol, catecholamines). Chronic stress increases their proliferation rate [46]. But proliferation alone does not explain the chromatin specificity. Our model adds: the chromatin remodeling machinery in these cells is under elevated demand because systemic stress forces context switches (changes in gene expression program) at a rate that exceeds the repair capacity for these specific loci.

9.2 The testable prediction

If context-switching load is the driver, then:

1. **Chromatin mutations should be increasing faster than other mutation types.** Cancer registry data with temporal resolution (SEER, AACR GENIE, TCGA vs. modern panels) should show that the fraction of tumors with chromatin remodeling mutations is increasing over time, independent of improved detection.

2. **The increase should be strongest in younger populations.** Digital natives (born after 1995) have the highest lifetime context-switching load. If the mechanism is correct, they should show earlier onset of CHIP and elevated chromatin mutation rates relative to age-matched historical controls.

3. **CHIP should appear at younger ages in high-information-exposure populations.** Tech workers, social media professionals, and other high-context-switching occupations should show earlier CHIP onset than low-context-switching occupations, after controlling for traditional risk factors.

4. **The increase should NOT be concentrated in known carcinogen-signature genes.** Environmental carcinogen exposure produces characteristic mutational signatures (e.g., C>A transversions from tobacco, CC>TT from UV). If the mutation increase is driven by information exposure rather than chemical exposure, the new mutations should lack these signatures and instead show the replication-error signatures associated with DNA break repair.

The last prediction is the discriminating test. Carcinogen-driven increases produce signature-specific mutations in genes throughout the genome. Context-switching-driven increases produce repair-error mutations concentrated in chromatin remodeling genes. These are distinguishable in sequencing data.

9.3 Cancer incidence trends

Cancer incidence in adults under 50 has been increasing globally, a trend termed "early-onset cancer" [47]. The increase is not fully explained by improved detection or known risk factors. The cancers with the largest increases include colorectal, breast, and pancreatic, all of which have high rates of chromatin remodeling gene mutations.

We do not claim that information overload is the sole driver. But we predict that it is a contributor, operating through the specific mechanism described here: elevated context-switching load →

elevated DNA break rate in chromatin remodeling loci → earlier exhaustion of repair capacity → earlier onset of somatic mutations in the genes that control cellular context → earlier cancer.

This is testable. It requires longitudinal cfDNA data from cohorts stratified by information exposure, with mutation calling focused on chromatin remodeling genes and mutational signature analysis. The prediction is specific: chromatin remodeling VAF should correlate with context-switching load after controlling for age, carcinogen exposure, and family history.

10 Discussion

The central claim of this paper is simple. Learning breaks DNA. Inefficient learning breaks more DNA per bit learned. Psychiatric disorders that reduce learning efficiency increase the ratio of DNA breaks to knowledge acquired. The genes most affected are the chromatin remodeling genes that control context switching at the molecular level. These are the same genes whose somatic mutation drives cancer.

This creates a direct, testable, molecular link between the efficiency of cognition and the risk of cancer: not through stress hormones, not through behavioral factors, not through immune suppression, but through the physical cost of computation in a biological substrate.

The inverted-U prediction distinguishes this framework from simpler "stress causes cancer" models. Stress models predict monotonically increasing cancer risk with increasing psychiatric severity. We predict the opposite at the extreme: the most severe context failures are protective because they disable the machinery that cancer needs.

The clinical implication is that treating context window disorders is not merely palliative. It is potentially disease-modifying for cancer. The three-tier protocol, interrupt, optimize, tune, is ordered by the logic of the framework: you cannot optimize a scheduler that is currently thrashing. Halt the runaway process first.

The deepest implication is for the relationship between knowledge and biology. If learning has a DNA damage cost, and if that cost is specific to the genes that control gene expression context, then what we know is physically inscribed in the error pattern of our DNA repair. Not in the sequence itself, like the immune system's somatic hypermutation, but in the *wear pattern* on the machinery that controls which sequences are readable.

If the mechanism proposed here holds, culture does not merely select on existing genetic variation. It writes on us, not as metaphor, but as somatic mutation in chromatin remodeling genes, accumulated across a lifetime of context switches, at a rate determined by how efficiently we learn. This is the speculative endpoint of the argument, but it is falsifiable: Prediction 12 provides the test.

Conflict of Interest

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References

[1] P. D. McCarthy, "Graph structure is necessary for information preservation under bounded context," 2026.

[2] P. D. McCarthy, "Evaluation-driven descent, encoding permanence, and the structural invariants of intelligence," 2026.

[3] P. D. McCarthy, "Coupling-channel count predicts cancer survival better than mutation count," 2026.

[4] P. D. McCarthy, "Determinism precipitates from uncertainty: when agents should become scripts," 2026.

[5] R. Madabhushi et al., "Activity-induced DNA breaks govern the expression of neuronal early-response genes," *Cell*, vol. 161, pp. 1592–1605, 2015.

[6] T. Lindahl, "Instability and decay of the primary structure of DNA," *Nature*, vol. 362, pp. 709–715, 1993.

[7] S. Sun et al., "Association of attention-deficit/hyperactivity disorder with cancer risk," *JAMA Network Open*, vol. 5, e2249253, 2022.

[8] Y.-H. Chen et al., "Increased risk of brain cancer in people with traumatic brain injury," *Br. J. Cancer*, 2012.

[9] J. L. Gradus et al., "Posttraumatic stress disorder and cancer risk," *JAMA Psychiatry*, vol. 72, pp. 1179–1185, 2015.

[10] C.-C. Shen et al., "Risk of cancer in patients with obsessive-compulsive disorder," *J. Clin. Psychiatry*, 2013.

[11] R. Tao et al., "Association between behavioral addiction and cancer risk," *Psycho-Oncology*, 2017.

[12] Y. Chida et al., "Does psychosocial well-being predict cancer incidence?" *Nature Rev. Clin. Oncol.*, vol. 5, pp. 466–475, 2008.

[13] G. D. Batty et al., "Psychosocial factors and cancer incidence," *BMJ*, 2017.

[14] E. E. McGinty et al., "Cancer incidence in a sample of Maryland residents with serious mental illness," *Psychiatric Services*, vol. 63, pp. 714–717, 2012.

[15] V. S. Catts et al., "Cancer incidence in patients with schizophrenia," *Schizophrenia Research*, vol. 102, pp. 137–162, 2008.

[16] J. Ji et al., "Cancer risk in schizophrenia," *BMC Cancer*, vol. 13, 2013.

[17] T. Singh et al., "Rare schizophrenia risk variants are enriched in chromatin remodeling genes," *Nature Genetics*, vol. 48, pp. 400–404, 2016.

[18] M. Solmi et al., "Cancer risk in antipsychotic-treated schizophrenia," *Lancet Psychiatry*, 2020.

[19] J. A. Driver et al., "Inverse association between cancer and Alzheimer's disease," *BMJ*, vol. 344, e1442, 2012.

[20] M. Musicco et al., "Inverse occurrence of cancer and Alzheimer disease," *Neurology*, vol. 81, pp. 322–328, 2013.

[21] M. I. Behrens et al., "Inverse relationship of Alzheimer's disease and cancer," *JAMA Neurol.*, 2009.

[22] B. W. Darbro et al., "Autism linked to increased oncogene mutations but decreased cancer rate," *PLOS ONE*, vol. 11, e0149041, 2016.

[23] Y. Lu et al., "Insomnia and cancer risk," *Sleep Medicine Reviews*, 2017.

[24] A. Bajaj et al., "Parkinson's disease and cancer risk," *Cancer Causes Control*, vol. 21, pp. 697–707, 2010.

[25] H. Hasle et al., "Cancer incidence in Down syndrome," *Lancet*, vol. 355, pp. 165–169, 2000.

[26] K. Patja et al., "Cancer incidence of persons with intellectual disability," *J. Intellect. Disabil. Res.*, 2001.

[27] G. W. E. Santen et al., "Mutations in *ARID1A* cause intellectual disability," *Am. J. Hum. Genet.*, 2012.

[28] N. Miyake et al., "*KMT2D* and Kabuki syndrome," *Am. J. Med. Genet.*, 2013.

[29] Y. Tsurusaki et al., "Mutations in *SMARCA4* cause Coffin-Siris syndrome," *Nature Genetics*, 2012.

[30] K. Tatton-Brown et al., "Mutations in *DNMT3A* cause Tatton-Brown-Rahman syndrome," *Am. J. Hum. Genet.*, 2014.

[31] C. Kandoth et al., "Mutational landscape and significance across 12 major cancer types," *Nature*, vol. 502, pp. 333–339, 2013.

[32] J. C. Wan et al., "Liquid biopsies come of age," *Nature Reviews Cancer*, vol. 17, pp. 223–238, 2017.

[33] J. K. Kiecolt-Glaser et al., "Psychoneuroimmunology and cancer," *Ann. Oncol.*, 2002.

[34] N. Thadathil et al., "DNA double-strand breaks accumulate in aging brain," *Frontiers in Aging Neuroscience*, 2021.

[35] C. H. Waddington, "Canalization of development and the inheritance of acquired characters," *Nature*, vol. 150, pp. 563–565, 1942.

[36] D. Church et al., "The effect of emotional freedom techniques on stress biochemistry," *J. Nerv. Ment. Dis.*, vol. 200, pp. 891–896, 2012.

[37] N. S. Schutte and J. M. Malouff, "A meta-analytic review of the effects of mindfulness meditation on telomerase activity," *Psychoneuroendocrinology*, 2020.

[38] J. D. Creswell et al., "Alterations in resting-state functional connectivity link mindfulness meditation with reduced interleukin-6," *Biol. Psychiatry*, 2016.

[39] J. Chubak et al., "Antidepressant use and colorectal cancer risk," *Cancer Causes Control*, 2011.

[40] S. Dalsgaard et al., "ADHD medication and cancer risk," *J. Child Psychol. Psychiatry*, 2018.

[41] T. I. Barron et al., "Beta-blockers and breast cancer mortality," *J. Clin. Oncol.*, vol. 29, pp. 2635–2644, 2011.

[42] C.-H. Kao et al., "Benzodiazepine use and risk of cancer: a population-based cohort study," *Psycho-Oncology*, 2017.

[43] A. L. Franks and J. E. Slansky, "Autoimmune disease and cancer," *Autoimmunity*, 2012.

[44] S. Jaiswal et al., "Age-related clonal hematopoiesis associated with adverse outcomes," *N. Engl. J. Med.*, vol. 371, pp. 2488–2498, 2014.

[45] D. P. Steensma et al., "Clonal hematopoiesis of indeterminate potential and its distinction from myelodysplastic syndromes," *Blood*, vol. 126, pp. 9–16, 2015.

[46] T. Heidt et al., "Chronic variable stress activates hematopoietic stem cells," *Nature Medicine*, vol. 20, pp. 754–758, 2014.

[47] H. Sung et al., "Global trends in incidence of early-onset cancer," *BMJ Oncology*, 2024.